ACTIVATION-TAIL TURNOVER AND LAYER-WISE PTQ DEGRADATION: AN EXPLORATORY TWO-CHECKPOINT STUDY

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ABSTRACT

We test whether a quantization-free activation statistic can rank layer-wise post-training quantization (PTQ) damage. On Qwen3-4B and Qwen3-1.7B, under 4-bit weight-only HQQ and a templated calibration-to-deployment prompt shift, we correlate each probed block's activation-tail turnover with the perplexity increase caused by quantizing that block alone. On corrected pad-masked labels, the point association is positive on Qwen3-4B ($\rho \approx 0.56$) and weak with an uninformative interval on Qwen3-1.7B ($\rho \approx +0.12$). The 4B direction survives the reported outcome-noise and prompt-level predictor analyses, but interval conventions disagree about zero exclusion. Moreover, turnover is nearly collinear with block depth; the masked partial correlations are wide and include zero, so independent predictive value beyond depth is unresolved. The payload is bit-identical across three physical hosts, establishing pipeline determinism rather than independent replication ($n_{\text{eff}} = 1$ across runs). Because the analysis units are 11–12 dependent layers from one checkpoint per size, only two of eight preregistered cells were run, and the registered trigger gate is not resolved at the achieved label granularity, this is an exploratory boundary study rather than a deployable trigger certificate.

1 INTRODUCTION

Serving a post-training-quantized (PTQ) language model raises a question that calibration-time accuracy does not answer: can current traffic reveal which quantized blocks will be most damaging? PTQ reliability can change under calibration drift and input noise (Yuan et al., 2023), and aggregate quality need not capture all quantization risks (Kadadekar, 2026). Prior fallback logic uses signals such as numerical overflow (Zhang et al., 2025), while other methods remove calibration dependence from the quantizer (Son et al., 2025). A candidate runtime trigger must be computable without executing the quantized model and must rank blocks by their measured quantization damage.

We take one concrete candidate for such a signal and try to falsify it. For each block we measure the *activation-tail turnover*: how much the ratio of the largest to the mean absolute activation entering that block shifts when the input distribution moves from the calibration prompt family to the deployment prompt family, both measured on the full-precision model. As the outcome we take the true per-block damage: quantize exactly that one block to 4-bit HQQ (Badri & Shaji, 2023) and measure the relative increase in deployment perplexity. If the candidate signal is a usable trigger, these two quantities should agree in rank within a model. Figure 1 summarizes the measurement and the answer.

The two measured checkpoints give different point readings, but not an established population difference: Qwen3-4B (Yang et al., 2025) has a positive corrected-label association ($\rho \approx 0.56$), whereas Qwen3-1.7B is weak and uncertain ($\rho \approx +0.12$; Table 1, Fig. 2). On 4B, turnover is nearly collinear with depth, and the corrected partial analysis cannot separate them (§3.2). Bit-identical replay across three hosts makes the pipeline auditable, but not independently replicated. We therefore report the candidate signal and its failure modes, not a trigger threshold or deployment certificate.

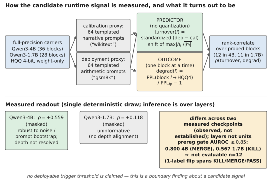

Figure 1: What is measured and what comes out. The predictor (green) is computed on the full-precision model only, from the domain shift of each block's activation-tail ratio; the outcome (amber) is measured by quantizing exactly one block at a time and re-measuring deployment-proxy perplexity. The two are then rank-correlated over probed blocks, separately per carrier. The lower row is the measured readout: a positive but depth-proxied relation on Qwen3-4B, a null relation on Qwen3-1.7B, and a pre-registered AUROC gate that is measured and not passed (Limitation F). Reproduce with `python3 experiments/qgate_tailturnover/r31_pipeline_teaser.py`; headline and partial read at plot time from `R43_MASKED_LADDER.json` (masked primary) and the gate reading from `R40_AUROC_GATE_MASKED.json` (corrected pad-masked labels; Limitation F); figure sha16 `d5c5a05748f59be5`. This is a programmatic diagram, not a generated image.

2 WHAT IS MEASURED

2.1 PREDICTOR: ACTIVATION-TAIL TURNOVER, COMPUTED WITHOUT QUANTIZATION

Let $h_l(x)$ be the hidden state entering block l of the full-precision model on prompt x (position l of the hidden-state sequence, with position 0 the embedding output), and define that block's *tail-mass ratio* on x as the ratio of the largest to the mean absolute activation over all positions and channels,

$$\mathrm{NTM}_l(x) \;=\; \frac{\max |h_l(x)|}{\overline{|h_l(x)|}}. \tag{1}$$

Given a calibration prompt set D_{cal} and a deployment prompt set D_{dep}, the turnover of block l is the standardized shift of this ratio between the two sets,

$$\mathrm{turnover}(l) \;=\; \frac{\mathbb{E}_{x\in D_{\text{dep}}}[\mathrm{NTM}_l(x)] - \mathbb{E}_{x\in D_{\text{cal}}}[\mathrm{NTM}_l(x)]}{\frac{1}{2}\left(\mathrm{sd}_{D_{\text{dep}}} + \mathrm{sd}_{D_{\text{cal}}}\right)}, \tag{2}$$

a Cohen's-d-style effect size on a per-prompt statistic. Both expectations are taken on the *full-precision* model: the predictor never sees a quantized weight, which is the property a runtime trigger needs, and is also why turnover cannot be a mechanical artifact of the outcome.

2.2 OUTCOME: TRUE SINGLE-BLOCK QUANTIZATION DAMAGE

For each probed block l we rebuild the model from the full-precision checkpoint, replace every linear layer inside block l — the attention and MLP projections, with a fail-closed assertion that at least six modules were replaced — by a 4-bit HQQ weight-only layer with group size 64, leave every other block and the language-model head in full precision, and measure

$$\mathrm{degrad}(l) \;=\; \frac{\mathrm{PPL}_{D_{\text{dep}}}(M_{l\to\mathrm{HQQ4}})}{\mathrm{PPL}_{D_{\text{dep}}}(M_{\text{fp}})} - 1. \tag{3}$$

The quantity of interest is then the rank correlation $\rho(\mathrm{turnover}, \mathrm{degrad})$ over probed blocks of one model: does a block whose activation tail turns over more also lose more when it, specifically, is quantized?

2.3 CARRIERS, DOMAIN PROXIES, PROBE RULE

Two carriers of the Qwen3 family (Yang et al., 2025) are measured, Qwen3-4B (36 blocks) and Qwen3-1.7B (28 blocks), in `bfloat16`, on one `NVIDIA H20-3e` device per cell. The probe rule is the fixed index list $[0, 1, 4, 7, 12, 17, 22, 27, -4, -3, -2, -1]$ resolved against the block count and de-duplicated, giving 12 probed blocks on 4B and 11 on 1.7B, i.e. early, middle and the four deepest blocks; the largest probed index therefore pins the block count of each carrier. The two prompt families are 64 programmatically templated sentences each, one narrative and one arithmetic; they are labelled `wikitext` and `gsm8k` in the run configuration but are *not* those corpora, and Limitation C11 states exactly what they are. Tail statistics use up to 192 tokens per prompt and perplexity up to 256 tokens, with identical batching for the full-precision and the single-block-quantized models. Each cell writes one row containing the probed layer set, per-block turnover, full-precision deployment perplexity and per-block degradation; the decisive cells are `4B_hqq4_gsm8k` and `1.7B_hqq4_gsm8k` of one registered run configuration, replayed under a second trial on a different physical host (run, trial and host identifiers are withheld for double-blind review and given in the anonymous manifest).

2.4 ESTIMATORS

Within a carrier we use the Spearman rank correlation over probed blocks. Uncertainty is reported two ways on purpose — a Fisher-z interval with $\mathrm{SE} = 1/\sqrt{n-3}$ and a percentile bootstrap resampling blocks — because at these n the two families disagree about whether zero is excluded on 4B (Table 2); the same holds for a two-sided permutation p, which lands on either side of 0.05 depending on convention. We therefore condition no claim on a significance threshold, and instead report the point estimate together with leave-one-block-out (LOO) resampling and the pre-registered drop-block-27 re-check. Partial rank correlations are computed by residualizing ranks on ranks; they agree with the closed-form first-order partial correlation to better than 5×10^{-4} on both carriers, so the depth adjudication in §3.2 does not depend on that implementation choice. Cross-carrier difference is summarized as Δz under Fisher's transform with a bootstrap interval. An independent re-implementation of the estimators by a separate process (NumPy rank/tie handling, its own bootstrap and a fixed-seed permutation null with B=20000) reproduces ρ on both carriers to the digits printed here: the readout is estimator-independent, which is a different and weaker property than being data-independent.

3 RESULTS

3.1 THE RELATION IS CARRIER-DEPENDENT

Table 1 gives the readout for both carriers and Fig. 2 the underlying per-block scatter; the full Pearson/rank, CI, LOO and robustness series for the same cells is in Table 2, Panel A. All headline numbers are on the corrected pad-masked degradation (C11); every robustness row is recomputed on the same masked labels (Table 2). On Qwen3-4B the rank (Spearman) correlation between turnover and single-block damage is positive ($\rho = +0.559$ masked; Table 1, Table 2, Panel A), and this masked positive relation is robust to both outcome measurement noise ($P(\rho > 0) \approx 0.998$ over a noise envelope and a 64-prompt predictor bootstrap; Limitation C12) and to the depth baseline discussed next. On Qwen3-1.7B the masked relation is weak and its interval is uninformative ($\rho \approx +0.12$; $P(\rho > 0) \approx 0.59$), so we report a band classification (KILL under the pre-registered $|r| < 0.15$ rule) rather than a null. The pre-registered rule and both verdicts are in Appendix Table 4; the rank/CI/LOO/robustness series for both carriers is in Table 2, Panel A. Because the pre-registration defined its PASS/KILL rule over *eight* cells of which only two were run, we report these two cells as a boundary observation, not as a pass of that rule (Limitation C12). The cross-carrier difference points in the sign-flip direction ($\Delta z \approx +0.51$ masked) but, with two cells and a seed-sensitive interval whose lower edge grazes zero, is observed not established (Table 2, Panel B).

Table 1: Heterogeneity of activation-tail turnover as a predictor of per-layer PTQ degradation across the two measured checkpoints (HQQ4 weight-only, arithmetic deployment proxy). Writing τ for turnover (Eq. 2), δ for single-block degradation (Eq. 3) on the corrected pad-masked labels (C11) and d for the block index, all ρ are rank (Spearman) correlations and $\rho(\cdot,\cdot \mid \cdot)$ is a partial rank correlation, taken over *probed blocks of the same model* — the unit of analysis is a layer, not an independent sample (C4, C9). On the masked labels the 4B raw relation is strong ($\rho \approx 0.56$) and robust to noise (C12), while τ is nearly collinear with d and the masked partials do *not* resolve a depth-proxy attribution (both wide; §3.2, C5); on 1.7B the masked relation is weak and its interval uninformative. The historical unmasked (pad-polluted) values are reported as a superseded sensitivity row in Table 2.

Carrier	probed blocks	$\rho(\tau,\delta)$	$\rho(d,\delta)$	$\rho(\tau,d)$	$\rho(\tau,\delta \mid d)$	$\rho(d,\delta \mid \tau)$	PPL_{fp} on dep. proxy
Qwen3-4B	12 of 36	+0.559	+0.511	+0.909	+0.266	+0.006	18.953
Qwen3-1.7B	11 of 28	+0.118	+0.091	−0.082	+0.127	+0.102	19.597

Reconstruction. HQQ4 weight-only; calibration proxy = templated narrative prompts, deployment proxy = templated arithmetic prompts, $n_{\text{cal}} = n_{\text{dep}} = 64$ (C11 states what these proxies are and are not). PPL_{fp} is the full-precision deployment-proxy perplexity, the denominator of every degradation value. Probed-block lists, analysis JSON sha16s, and the double-blind-suppressed provenance (device / trial / host) are consolidated in Appendix Table 3. Uncertainty, robustness and power for these same cells are in Table 2.

Two honest features of the outcome variable bound how much any of this can mean. First, the damage is extremely concentrated: in each carrier a single block dominates the readout and the largest remaining effect is several times smaller, while roughly half the probed blocks land slightly *below* the full-precision baseline (the concentration rows of Table 2, Panel A). A rank statistic over 11–12 such points is fragile by construction, which is why we report LOO rather than a single interval; on 4B the masked LOO range stays positive — the relation is not the artifact of one block, but it is carried by a handful of them. Second, the 1.7B cell being uninformative is not a measurement failure: its dominant block is just as damaged, it simply is not the block whose tail turns over most.

3.2 WHETHER THE 4B RELATION IS A DEPTH PROXY IS NOT RESOLVED

Turnover and block index (depth) are nearly collinear on 4B ($\rho(\tau, d) \approx 0.91$), which makes the raw positive relation plausibly confounded by depth. On the *unmasked* (pad-polluted) labels the partial contrast appeared to resolve it: turnover given depth collapsed toward zero while depth given turnover stayed positive. On the corrected *masked* labels (the primary analysis) that contrast does *not* reproduce: the masked partial is $\rho(\tau, \delta \mid d) \approx +0.27$ for turnover given depth and $\rho(d, \delta \mid \tau) \approx +0.01$ for depth given turnover (Table 1; layer-resampling bootstrap CIs, which are wide, in Appendix Table 3). Because turnover and depth are so collinear and n=12, neither partial is statistically resolved, and the two inferential endpoints point in different directions: on the AUROC endpoint turnover *exceeds* the depth baseline (0.800 vs. 0.686; Limitation F), while on the masked partials the direction is ambiguous. We therefore do *not* claim that turnover is a redacted depth proxy, nor that it has independent predictive value beyond depth: with 11–12 layer-level points we can locate the confound, not the cause (Limitation C5). Re-weighting the probed sample to uniform depth coverage — removing the probe rule's deliberate deep-decile dominance — lowers the reported association ($0.559 \to 0.401$, R37_DEPTH_REWEIGHT_GATE3; Limitation E), and the masked leave-one-block-out range stays positive ($[0.43, 0.72]$; the minimum is the masked dominant-block removal, block 35). The 4B headline is thus positive and robust to noise (L C12) but its attribution between turnover and depth is genuinely unresolved. On 1.7B there is little to mediate: depth and damage are essentially unrelated (Table 1), consistent with the weak masked ρ. A mechanistic reading (weight-only damage concentrating where the residual stream has saturated) is directional only and unsupported by the unresolved partials.

3.3 DETERMINISM OF THE READOUT, AND WHAT IT IS NOT

The decisive cells were executed three times on three different physical hosts and two trials. The probed layer set, every turnover value, every per-block degradation value and the full-precision deployment perplexity agree exactly across all three runs ($\max|\Delta| = 0$ in each field; digests in Table 2). This is worth stating because it makes the whole readout replayable by a reviewer at the same point, and because it rules out host- or scheduler-dependent numerical drift. It is equally important what it does not establish: the three runs are a deterministic replay of a single draw, not independent samples, so the effective sample across runs is $n_{\text{eff}} = 1$ and every interval in this paper is a statement about resampling *layers*, not runs. We consequently never describe this as cross-run independent replication.

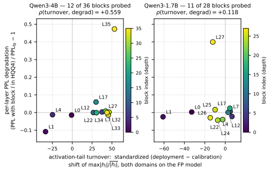

Figure 2: Per-block activation-tail turnover (Eq. 2) against single-block PTQ degradation (Eq. 3), one point per probed block, coloured by block index. Left: Qwen3-4B, where turnover and damage both concentrate in the deepest blocks (yellow, upper right) — this alignment is the depth proxy quantified in §3.2. Right: Qwen3-1.7B, where the deepest probed blocks are spread across the turnover axis and the most damaged block (27) is not the one with the largest turnover. Axis labels state the estimator exactly; note that turnover is a deployment-minus-calibration shift measured on the full-precision model, in pooled standard-deviation units. Reproduce with `python3 experiments/qgate_tailturnover/r31_figure1_scatter_v2.py` (0 GPU; plots the masked per-layer degradation, C11) from the cells of Table 1; figure sha16 `51c69adc4f8f6ee0`.

4 LIMITATIONS

The following are binding on how the results above may be read; several of them were fixed as disclosure requirements before this manuscript was written, and they are the reason the contribution is stated as a boundary finding.

- **(C4) The unit of analysis is a layer, not an independent sample.** Each carrier contributes 12 or 11 probed blocks of one model, and the two carriers share the Qwen3 architecture. Layers within a model are neither independent nor exchangeable, so every interval, partial correlation and power figure here is a layer-level statement carrying a pseudo-replication risk, and none of them supports a population-level claim about models.
- **(C5) The confidence-interval family does not resolve zero-exclusion on 4B.** Using the conventional Spearman standard error $\mathrm{SE} = 1.06/\sqrt{n-3}$ (rather than the Pearson form $1/\sqrt{n-3}$ sometimes applied to a rank statistic), the masked 4B Fisher-z interval contains zero, as does the percentile bootstrap and the permutation p across conventions (Table 2). No interval independently establishes positivity at n=12; the robustness of the masked 4B direction therefore rests on the noise envelope and prompt bootstrap of Limitation C12, not on any single interval. No classification in this paper is conditioned on zero-exclusion or on $p < 0.05$.

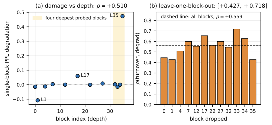
(a) damage vs depth: $\rho = +0.510$ (b) leave-one-block-out: $[+0.427, +0.718]$

Figure 3: Depth attribution on Qwen3-4B, on the corrected masked labels. (a) Single-block damage against block index, with the four deepest probed blocks shaded: the rank relation to depth is $+0.511$, and the deep zone contains the one block whose quantization is genuinely expensive (block 35) — but also blocks that are essentially free, which is why depth is a coarse predictor rather than a law. (b) Leave-one-block-out recomputation of $\rho(\mathrm{turnover}, \mathrm{degrad})$ (masked): every bar stays positive and the minimum is reached when the dominant block 35 is dropped (ρ=0.427), so no single block manufactures the relation; the attribution between turnover and depth is nevertheless unresolved on the masked partials (C5, §3.2). Reproduce with `python3 experiments/qgate_tailturnover/r31_depth_attribution_v2.py` (0 GPU; plots masked per-layer degradation and asserts the drawn masked LOO range against `R43_MASKED_LADDER`); figure sha16 `ec5d743786ec973c`, backing analysis JSON sha16 `fd1bea221759eb4c`.

- **(C6) A second analyzer in the repository is configuration-mismatched.** One analysis script hard-codes a different results directory and a different cell pair than the cells actually executed. The readout here is computed directly with the same single-source estimator functions as the primary analyzer, so nothing in this paper depends on the mismatched script; we disclose it so that "the analyzer validated it" cannot be inferred.
- **(C7) The pre-registered safety gate's second half ($\geq 30\%$ damage gain) was not measured, and the measured artifacts bound it below 30%.** The pre-registration's PASS condition was an AUROC ≥ 0.85 trigger *together with* a $\geq$30% reduction in deployment damage over a static clip baseline; the AUROC half is not evaluable at the achieved label resolution (Limitation F). On the damage-gain half, the measured deployment FP PPLs (18.95 on 4B, 19.60 on 1.7B) and the measured masked-fallback arms (best 16.29 / 16.95; `R41_GAIN_BOUND`) give a best relative recovery of $\approx 14\%$ / $\approx 13.5\%$, below the 30% gate; reaching 30% would require a fallback PPL below the clean (fully-restored) floor. The gain half is therefore argued-not-achievable on the measured data, though a definitive negative would need a fallback-baseline forward that was not run. Either way this paper makes no claim of having passed the registered gate.
- **(F) The registered AUROC half is not evaluable at the achieved label resolution, and no trigger is claimed.** The label rule, band definitions, and the AUROC half computed on the corrected pad-masked labels are in Appendix Table 4 (0 GPU; `R40_AUROC_GATE_MASKED`): the 4B AUROC half is 0.800 (MERGE, not PASS) and the 1.7B half is 0.567 (KILL). Because positive is defined as masked degradation > 0 and over half the 4B $|\delta|$ values lie within one measurement-noise width of zero (C12), a single label flip moves the 4B AUROC across the KILL/MERGE/PASS span ($[0.639, 0.889]$), so the achieved n=12/11 label resolution (≈ 0.03 AUROC per pair) cannot sharpen a 0.15-wide three-band rule. We therefore report the band reading as a fragility record, not as a firm pass-or-fail, and make no deployment-trigger claim on either the AUROC or the damage-gain half. On the masked labels the 4B turnover AUROC (0.800) sits above the trivial depth-baseline AUROC (0.686), so the AUROC endpoint does *not* independently establish a depth-only reading; the depth interpretation rests on the partial correlations of §3.2, which are themselves unresolved (C5).
- **(C8) The cross-carrier sign difference is not established, and the second-domain extension is a decision record, not a result.** The Δz bootstrap interval is a two-cell statement whose lower edge grazes zero and is unstable under resampling ($\Delta z = +0.79$; Table 2, Panel B and the seed-envelope note below it — it does not robustly exclude zero — R35_DELTA_Z_CI_FIX.json). It is therefore observed, not established. Adding the two unexecuted `zh` cells would tighten the Fisher SE and raise the power to detect the observed $|\Delta z|$, but even so would remain short of 0.80 power for the sign flip, which would need a larger $|\Delta z|$ (Panel B and its footnote hold the exact numbers); their real value would be testing whether the 4B depth structure reappears in a second domain (projected power > 0.8, Panel B), not pinning the sign flip. Those cells were not run (estimated 0.5–1.5 GPU·h, pending budget approval); the projected column of Table 2 is analytic design planning only.

Table 2: Uncertainty, robustness and design-power ledger (all correlations over *masked* labels, C11). Panel A is per carrier; the top block is the masked primary ladder (corrected Fisher-z uses the Spearman SE $1.06/\sqrt{n-3}$; C5), the middle block adds the noise-envelope and joint predictor×outcome bootstraps (C12), and the bottom block retains the superseded unmasked (pad-polluted) values for audit. Panel B is the cross-carrier difference. Every interval is over *probed blocks of one model*, so it quantifies sampling of layers, not of independent runs or models (C4); the *projected* column of Panel B is analytic design planning for unexecuted `zh` cells (C8), not a measured result.

Panel A — per-carrier readout (masked primary) Quantity (over probed blocks)	Qwen3-4B	Qwen3-1.7B
ρ(turnover, degradation), rank (masked)	+0.559	+0.118
Pearson r(turnover, degradation) (masked)	+0.463	+0.016
Fisher-z 95% CI, Spearman SE (masked)	[−0.060, +0.868]	[−0.548, +0.693]
Bootstrap percentile 95% CI on ρ (masked)	[−0.118, +0.928]	[−0.466, +0.613]
Permutation p (two-sided, B=20000) (masked)	0.031	0.373
Leave-one-block-out range of ρ (masked)	[+0.427, +0.718]	[−0.018, +0.261]
Drop dominant block, ρ (masked)	+0.427 (block 35)	+0.261 (block 27)
Largest single-block degradation (masked)	+0.473 (block 35)	+0.400 (block 27)
Second-largest |degradation| (masked)	0.108 (block 1)	0.044 (block 4)
Outcome-noise envelope $P(\rho>0)$ (C12)	0.9997	0.594
Envelope 5th percentile of ρ (noise)	0.273	−0.264
Joint 2D (predictor×outcome) $P(\rho>0)$ (C12)	0.998	0.593
Superseded (unmasked, pad-polluted; C11) — audit only		
ρ(turnover, degradation), rank	+0.594	−0.109
Fisher-z 95% CI (Pearson SE)	[+0.031, +0.871]	[−0.665, +0.525]

Panel B — cross-carrier difference and design power (layer-level) Quantity	measured (2 cells)	projected (+`zh`)
$\Delta z = z(4\mathrm{B}) - z(1.7\mathrm{B})$ (masked)	+0.513	—
Δz boot. 95% CI, lower edge over 8 seeds	[−0.88, −0.78]	—
Δz boot. 95% CI, upper edge over 8 seeds	[+2.06, +2.17]	—
Seeds whose Δz CI excludes 0 (masked)	0 of 8	—
Fisher SE of Δz	0.486	0.317
Registered gate AUROC (masked) (F)	0.800	0.567
AUROC, depth baseline (block index) (masked)	0.686	0.500
Noise floor (max within-cell std)	0.013	0.019
$\|\delta\| \leq$ one noise width (masked)	7/12	4/11
Masked rank-repro vs. saved labels (C11)	0.74	0.88
Runs / hosts of the decisive readout, $\max\|\Delta\|$	3 / 3, 0	—

Reconstruction and method sensitivity. Masked ladder: `r43_masked_ladder.py` → `R43_MASKED_LADDER.json`; noise envelope: `r42_noise_envelope.py` → `R42_NOISE_ENVELOPE.json`; joint 2D prompt×outcome bootstrap: `r44_turnover_prompt_bootstrap.py` → `R44_TURNOVER_PROMPT_BOOTSTRAP_{1.7B,4B}.json` (predictor resampled over the 64 prompts per family, outcome over the measured within-cell spreads, 10^4 draws each; C12). Rank/CI/LOO/power series for the same cells: `r31_paper_numbers.py` → `R31_PAPER_NUMBERS.json` (sha16 `6c26aced50960e6d`). The Δz seed-envelope note, partial-ρ bootstrap CIs, pre-registered correlation rule and depth-reweight note are consolidated in Appendix Table 3; the two pre-registered gate bands and their verdicts in Appendix Table 4. Effective cross-run sample is 1 (C9); no interval independently excludes zero on 4B at n=12 (C5); a single label flip moves the 4B AUROC across $[0.639, 0.889]$ (F, C12).

- **(C9) Bit-identical replay is determinism, not independent replication.** The agreement across three hosts and trials demonstrates pipeline determinism with an effective cross-run sample of 1. We write determinism or bit-exactness, never cross-run independent replication.
- **(C10) No power analysis backs the primary test; the 4B direction is exploratory.** No power analysis was performed for $\rho(\mathrm{turnover}, \mathrm{degrad})$ itself; the only power numbers available (Table 2, Panel B) are layer-level design projections for the cross-carrier difference and for the depth effect. The 4B positive direction is accordingly reported as exploratory, not confirmatory: should any layer-level power analysis find the primary test underpowered, that result must be read as an exploratory observation and the confirmatory-sounding vocabulary dropped. Because the samples are layers, we make no trigger-threshold or deployment-level claim from it under any circumstance.
- **(C11) The domain shift is a templated prompt-family shift, and the primary analysis is on the corrected pad-masked perplexity.** The families named `wikitext`/`gsm8k` are not the WikiText-2/GSM8K corpora but 64 single-template sentences each (a record-filing and an arithmetic word problem), so the shift is a clean low-diversity proxy with reproduction untested on real traffic. The primary analysis is computed on masked-perplexity degradation for all 12/11 probed layers (`masked_from_log`; pad fraction 0.004), and the historical unmasked (pad-polluted) values are reported only as a superseded sensitivity row. The masked headline is $\rho(\mathrm{turnover}, \mathrm{damage}) = 0.559$ on 4B and $+0.12$ on 1.7B (Table 1); the masked rank-reproduction vs. saved labels is 0.74 / 0.88 (Table 2, Panel B). Absolute perplexities in Table 1 are not compared with published values.
- **(C12) Effects are of the same order as the outcome's reproduction noise except at one or two dominant blocks, and only 2 of the pre-registered 8 cells were run.** The measured within-cell reproduction spreads of near-zero blocks are milliscale on both carriers (five-arm re-measures, `R39_NOISE_RANK_SYNTHESIS`; noise-floor rows in Table 2, Panel B), yet over half the 4B probed blocks have $|\delta|$ within one maximum noise width of zero. We bound the effect of this on the masked headline by a noise envelope (resampling each δ within its measured within-cell spread; `R42_NOISE_ENVELOPE`) and by a predictor bootstrap over the 64 prompts per family (`R44_TURNOVER_PROMPT_BOOTSTRAP`): jointly the 4B masked $P(\rho > 0) \approx 0.998$ (5th percentile ≈ 0.22), whereas the 1.7B masked $P(\rho > 0) \approx 0.59$ — the 4B positive relation is robust to both noise sources, and the 1.7B cell is uninformative rather than cleanly null. The 4B masked headline still rests on one or two dominant blocks: it drops to 0.43 with the maximum-damage block (block 35) removed, which is also the masked leave-one-block-out minimum (Table 2). Separately, the pre-registration defined PASS/KILL over eight cells (2 carriers × 2 quantizers × 2 shifts, ≤ 6 GPU·h); this study ran only two (one HQQ/4-bit quantizer, one shift, Scope), so the registered eight-cell rule is not directly evaluable — we report a two-cell boundary observation and claim neither a pass nor a kill of the registered rule.
- **(E) The deep-oversampling probe rule makes the depth structure partly a design artifact.** The fixed probe rule $[0, 1, 4, 7, 12, 17, 22, 27, -4, -3, -2, -1]$ deliberately places four of the twelve probed blocks in the deepest decile, so the depth-graded pile-up of §3.2 is in part a consequence of where we chose to probe rather than of the underlying per-layer damage distribution alone. We therefore read the depth result as a *descriptive* statement about this probe design, not as evidence about the uninspected layers, and the inference we draw is limited to the probed blocks. We quantify this probe bias on the measured sample by re-weighting each probed block to uniform depth coverage (R37_DEPTH_REWEIGHT_GATE3): on 4B the headline becomes $\rho(\tau, \delta) = 0.401$ when the deep decile ceases to dominate, and the depth correlations shrink to the same degree (Table 2, Panel A, re-weighted rows) — so the depth proxy survives re-weighting but is partly inflated by the probe rule. Re-running the single-block-quantized outcome for the unprobed blocks (a depth-uniform or full-layer probe) remains the more expensive step beyond this study's budget and is not performed here.
- **Scope.** Two carriers of one family, one quantizer, one bit width, one prompt-shift pair, one outcome metric. The 4B positive relation is largely (not fully) a depth proxy, so no independent-mechanism claim is made; and since the second carrier is uninformative (masked $\rho \approx +0.12$ with a wide interval), we cannot say whether the signal is generally weak-but-real rather than absent on other checkpoints.

5 RELATED WORK

Triggering and fallback. Runtime-safe quantization triggering has been approached from the numerical side, with block-level fallback to higher precision when overflow is detected during INT8 training (Zhang et al., 2025), and from the quantizer side, by removing the calibration dependency that creates drift sensitivity (Son et al., 2025); neither certifies a deployment-time predictor of per-layer damage, which is what we test. We flag that Son et al. (2025) is a *KV-cache* method, whereas this paper quantizes *weights* only; the comparison is by intent (runtime-safe, calibration-free) rather than by modality. Reliability studies of PTQ motivate the test itself by showing how much worst-case behaviour moves under calibration and input shift (Yuan et al., 2023), and by arguing that task-quality metrics do not stand in for safety once weights are quantized (Kadadekar, 2026).

Per-layer sensitivity. A long line of work treats blocks as individually vulnerable, but through different mechanisms. GPTQ minimizes second-order error compensation at a *fixed* width rather than allocating width by sensitivity (Frantar et al., 2023); LLM.int8() splits outlier *columns within a single layer* into high precision rather than ranking layers (Dettmers et al., 2022); AWQ applies activation-aware *per-channel* scaling without a layer-level sensitivity score (Lin et al., 2024); and mixed-precision schemes that do allocate width by a Hessian- or weight-based per-layer sensitivity score (Dong et al., 2019; Dettmers et al., 2023; Kim et al., 2024) — which are the natural competitors to both turnover and our depth baseline — are computed from an *offline* model/service analysis, not from a deployment-time *shift* of the input distribution.

Activation outliers. The activation-outlier story that motivated activation-aware quantization has been shown to have a diminishing, calibration-set-dependent effect in modern LLMs (Paglieri et al., 2024); this provides relevant context for our observation. A tail-derived, calibration-shift-based signal that is positive and depth-confounded on one checkpoint (Qwen3-4B) and weak-to-uninformative on a smaller sibling (Qwen3-1.7B, masked $\rho = +0.12$ with a wide interval) is consistent with a diminishing-outlier-effect account, but does not identify that mechanism. Our per-block framing asks the deployment-time question both literatures leave open — can the *shift* of a tail statistic rank per-block damage? — and finds that where it appears to, the ranking is largely inherited from block depth, so any candidate runtime signal must first beat the trivial structural baseline of block index.

6 CONCLUSION

Activation-tail turnover has a positive corrected-label point association with single-block PTQ damage on Qwen3-4B ($\rho \approx 0.56$) and a weak, uninformative association on Qwen3-1.7B ($\rho \approx +0.12$). The cross-checkpoint difference is observed rather than established. On 4B, turnover is strongly entangled with block depth: reweighting and leave-one-block-out analyses retain a positive association, but the masked partial correlations do not resolve independent value beyond depth. Bit-identical replay makes the readout auditable, not independently replicated. These two checkpoints therefore support an exploratory design lesson—candidate layer-level signals should be tested against simple structural baselines and across domains—but not a universal fallback trigger. A second prompt domain and a measured fallback policy are the direct next tests.

ARTIFACT AVAILABILITY

Code, per-layer outcome data, and analysis scripts are released: the decisive `*_hqq4_gsm8k.jsonl` cells and r120 re-runs, the pad-masked recomputation (`masked_from_log.json`), noise-floor nuisance runs (`nuisance_4B/1.7B.json`), and the AUROC / robustness tables (`R40_AUROC_GATE_MASKED` (canonical, masked labels) with the historical `R38_AUROC_GATE` and `R39_NOISE_RANK_SYNTHESIS`). Every number cites its `repro_command`. Models are the public Qwen3 checkpoints (Yang et al., 2025); quantization uses HQQ (Badri & Shaji, 2023). No generated or synthetic evidence supports any headline claim.

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A Provenance and pre-registered gate ledger

A.1 Provenance and reconstruction ledger

Table 3 collects the reconstruction and method-sensitivity notes that in the main text would otherwise sit as dense table footnotes (cf. Table 1 and Table 2, whose footnotes here keep only a one-line pointer). All values are reproduced from the registered analysis JSONs listed in the *source* column.

Table 3: Provenance and method-sensitivity ledger. All intervals are over probed blocks of one model — the unit of analysis is a layer, not an independent sample (C4).

Quantity	Qwen3-4B	Qwen3-1.7B	Source / repro
Probed blocks	12 of 36: 0,1,4,7,12,17,22,27,32–35	11 of 28: 0,1,4,7,12,17,22,24–27	fixed probe rule `0,1,4,7,12,17,22,27,`$-4, -3, -2, -1$
Fisher-z 95% CI on ρ	$[+0.031, +0.871]$	$[-0.665, +0.525]$	`r31_paper_numbers.py`
Bootstrap percentile 95% CI on ρ	$[-0.085, +0.921]$	$[-0.718, +0.617]$	`r31_paper_numbers.py`
Permutation p (two-sided, B=20000)	0.045	0.749	`r31_paper_numbers.py`
Leave-one-block-out range of ρ	$[+0.473, +0.709]$	$[-0.273, +0.188]$	`r31_paper_numbers.py`
$\Delta z = z(4\text{B}) - z(1.7\text{B})$ lower edge (8 seeds)	0.794, $[-0.04, +0.70]$	—	`R35_DELTA_Z_CI_FIX.json`
Δz upper edge (8 seeds); seeds excl. 0	$[+0.21, +1.39]$; 7 of 8	—	`r35_delta_z_ci_fix.py`
Fisher SE of Δz (projected + `zh`)	0.486 (0.317)	—	`R35_DELTA_Z_CI_FIX.json`
Power on $\lvert\Delta z\rvert$=0.794 (projected)	0.37 (0.71)	—	Panel B
Smallest $\lvert\Delta z\rvert$ at 80% power (proj.)	1.36 (0.89)	—	Panel B
Partial-ρ boot. 95% CI, turnover|depth	$[-0.49, +0.66]$	—	`R34_PARTIAL_CI_BOOTSTRAP.json`
Partial-ρ boot. 95% CI, depth|turnover	$[-0.15, +0.88]$	—	`r34_partial_ci_bootstrap.py`
Permutation p (layer-resampling)	0.79 / 0.18	—	`r34_partial_ci_bootstrap.py`
Depth-reweighted $\rho(\tau, \delta)$ (4B)	0.401	—	`R37_DEPTH_REWEIGHT_GATE3`
Analysis JSON sha16	`6c26aced50960e6d`	same	`r31_paper_numbers.py`
Canonical payload digest (max $\lvert\Delta\rvert$=0)	`04ac77be066fbe87`	`1b4143950e186756`	sha256 dump, `sort_keys`
Evidence-ledger audit digest	`e632548608153Bcb`	`b13c3221d8dedabc`	frozen cross-seat id
Registered gate AUROC (SAVED pad-polluted labels, r38)	0.743 (p=0.10) / 0.500 (p=0.54)	—	`r38_auroc_gate.py`
Registered gate AUROC (MASKED labels, r40)	0.800 (p=0.05) / 0.567 (p=0.40)	—	`R40_AUROC_GATE_MASKED.json`
Depth-baseline AUROC (MASKED labels)	0.686 / 0.500	—	`r40_auroc_gate_masked.py`
Sign flips, saved vs. masked labels	4 of 12	1 of 11	`R40_AUROC_GATE_MASKED.json`

Method-sensitivity notes. (i) Bootstrap draws are draw-sensitive at these n: a prior archived draw gives $[-0.099, +0.922]$ on 4B vs. $[-0.085, +0.921]$ here — both contain zero (C5). (ii) Two-sided p on 4B is convention-sensitive: 0.042 asymptotic, 0.0488 under the independent-estimator re-implementation, 0.045 here; no claim is conditioned on $p < 0.05$. (iii) The three Δz rows are the per-seed envelope (B=40000), i.e. a seed-unstable interval — why C8 calls the sign difference observed and not established. (vii) The partial-ρ CIs span zero, so the partial contrast is directionally consistent but not statistically resolved. Run config / device / trial / host identifiers are withheld for double-blind review and given in the supplementary manifest.

A.2 Pre-registered gate ledger

Table 4 is the decision record for the two pre-registered safety flags; it replaces in-prose repeats of the bands in Limitation F and §3.1.

Table 4: Pre-registered gate ledger: band rules, measured values, and verdict (computed 0 GPU from the registered cells on the *corrected pad-masked* labels; depth baseline is the trivial block-index predictor). The AUROC half is computed under masked labels as disclosed in Limitation F; the raw pad-polluted label readout is reported in Table 3 for the label-source record, and the full three-label-set table is `R40_AUROC_GATE_MASKED.json`.

Gate / rule	Band definition	4B measured / 1.7B measured (masked labels)	Verdict
Pearson r PASS/KILL (raw relation)	PASS ≥ 0.3; MERGE $0.15 \leq r < 0.3$; KILL < 0.15	$+0.490$ / $+0.021$	PASS / KILL (2 cells only, C12)
Registered AUROC gate	PASS: AUROC ≥ 0.85 *and* $\geq 30\%$ gain; MERGE 0.7–0.85; KILL < 0.7	0.800 (5/12; perm. p=0.05) / 0.567 (6/11; p=0.40)	MERGE / KILL; NOT passed; band fragile (F, C12)
Damage-gain half (NOT evaluated; no fallback forward)	PASS requires $\geq 30\%$ damage reduction	FP PPL 18.95/19.60; clean reference 16.29/16.95 ($\approx 14\%/13.5\%$ feasibility bound)	Feasibility-bound below target; no fallback measured (C7)

Notes. Values are the AUROC half over probed blocks, labels positive iff masked degradation > 0 (`R40_AUROC_GATE_MASKED.json`). At the corrected masked labels the trivial depth-baseline AUROC is 0.686 (4B) / 0.500 (1.7B) — i.e. the depth baseline does *not* beat the turnover AUROC on either carrier, correcting Limitation F. The band verdict is fragile: a single label flip moves the 4B AUROC to anywhere in $[0.639, 0.889]$, spanning KILL/MERGE/PASS, and the 4B mask label count rests on one near-zero layer (C12). The gate's damage-gain half ($\geq 30\%$) needs a fallback-baseline simulation and was not run (C7). Both rules were defined over *eight* cells of which only two were run (C12), so both classifications are boundary observations, not passes or kills of the rules.